

Devichand Budagam

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RESEARCH INTERESTS

I conduct fundamental research broadly centered on **evolutionary intelligence**, **sample efficiency**, **meta learning**, **continual learning**, **information theory**, and **interpretability** with the goal of developing models that match or surpass human levels of sample efficiency and adaptability and that can continuously evolve in response to ever-changing data environments

EDUCATION

Indian Institute of Technology Kharagpur

Dec 2021 – May 2026

Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering

- **Cumulative GPA:** 8.45/10.00

PUBLICATIONS

- Sandeep Mishra, Devichand Budagam, Anubhab Mandal, Bishal Santra, Pawan Goyal, Manish Gupta. "Router-Suggest: Dynamic Routing for Multimodal Auto-Completion in Visually-Grounded Dialogs" **EACL 2026 Industry Track - ORAL** [PDF]
- Devichand Budagam, Ashutosh Kumar, Mahsa Khoshnoodi, Sankalp KJ, Vinija Jain, and Aman Chadha. "Hierarchical Prompting Taxonomy: A Universal Evaluation Framework for Large Language Models Aligned with Human Cognitive Principles" **KDD workshop on Prompt Optimization 2025, AAAI 2025 CogSci-AI Neuro-Symbolic Bridge** [PDF]
- Devichand Budagam, Mahsa Khoshnoodi, Jibesh Patra, Ravid Shwartz-Ziv, Amit Sheth, Vinija Jain, Aman Chadha. "Reimagining Neurosymbolic AI through the lens of Cognitive Science: A Survey" [**Under Review**] **ACM Computing Surveys**

RESEARCH EXPERIENCE

Teaching Assistant, Professor Pawan Goyal & Professor Aritra Hazra

Kharagpur, India

IIT Kharagpur

July 2025 – May 2026

- I served as the teaching assistant for a Deep Learning course, where I taught PyTorch, Hugging Face, and core deep learning coding concepts, and helped create and grade assignments.
- I served as the teaching assistant for both the Compilers Theory and laboratory courses, responsible for evaluating student assignments.

Undergraduate Researcher, Professor Pawan Goyal & Dr. Manish Gupta

Kharagpur, India

IIT Kharagpur

May 2025 – July 2026

- Formalized multimodal inline auto-completion; benchmarked text-only vs multimodal VLMs on dialogue tasks; built a latency-performance router to select text-only or multimodal models by prefix complexity
- I focused on examining and characterising the geometry of the data during the unmasking process of diffusion language models, investigating how it interacts with different unmasking schedulers, and quantitatively estimating the errors accumulated across the separate channels corresponding to model and the scheduler
- I worked on demonstrating the sub-optimality of existing schedulers using eigen decomposition of data geometry, while simultaneously uncovering and establishing a near-optimality bound for our new scheduler, and interpreting the information flow through tokens for different unmasking schedulers

Research Fellow, Professor Amit Sheth

South Carolina, USA

Artificial Intelligence Institute of South Carolina

July 2024 – August 2025

- Investigated current knowledge-enhanced pre-trained language models (KEPLMs) through the lens of abstraction and hierarchical abstraction structures drawing parallels with human cognition
- Explored how human-aligned abstraction hierarchies can serve as a learning objective for pre-training KEPLMs to enhance reasoning and structured understanding
- Proposed several computational frameworks based on abstraction ability, aiming to improve KEPLMs ability

to generalize and perform multi-level reasoning

Research Internship, Professor Ahmed Sayad
Queen Mary University of London

London, United Kingdom
Mar 2024 – June 2024

- Developed FedAgent, a benchmark to evaluate federated learning algorithms for tool calling with different LLMs
- Conducted experiments to evaluate the performance and scalability of multiple agent clients using homogeneous and heterogeneous mixtures of LLM instances with existing Federated Learning algorithms
- Conducted experiments using the XLAM-function and Openfunctions with varied data heterogeneity strategies to model i.i.d data distributions

Research Internship, Professor Dmitrii L.Kaplun
Saint Petersburg Electrotechnical University

St. Petersburg, Russia
May 2023 – Jan 2024

- Created UFBA-425, one of the world's largest open-source dental datasets with 425 panoramic X-rays with human-annotated bounding boxes and polygons
- Developed OralBBNet, an advanced architecture merging algorithms of object detection and segmentation, achieving a Dice Score of 0.87 and a mAP50 Score of 0.95
- Evaluated the dataset against existing dental segmentation methods, demonstrating the dataset's robustness and OralBBNet's superior performance

Research Fellow, Professor M.Sabarimalai
IIT Palakkad

Palakkad, India
Jul 2023 – Dec 2023

- Processed the ShipsEar database with FFT and pre-emphasis to obtain Mel spectrograms for training AST
- Implemented MelGAN for synthetic audio generation and evaluated the data quality using MFCC distortion scores and perceptual assessment
- Achieved 89% accuracy in audio classification by training AST with original and synthetic data

WORK HISTORY

Eng, Artificial Intelligence 2
KLA

Chennai, India
July 2026 – Present

- Working with nanometer-scale semiconductor data for defect classification across multiple modalities and physical features.
- Working on agentic harness for semiconductor yield and metrology data analysis.

Machine Learning Engineer Internship
KLA

Chennai, India
May 2025 – July 2025

- managed a team of engineers to implement an agent framework, execute model fine-tuning, curate data, and publish a paper on the work at the company's internal annual conference
- Developed an agent-based architecture to automate the generation of workflows for yield analysis and defect analysis on chip wafers
- Developed an algorithm for multi-configuration prediction by combining K-means clustering with a latent text encoder to capture both contextual and statistical patterns in configuration data

Co-Founder, CTO
Meridian (Backed by Founders Inc.)

San Francisco, USA
July 2025 – July 2026

- Built a proactive intelligence OS that consolidates, automates, and explains knowledge work for finance and accounting teams
- I led a team of eight people and successfully scaled the platform to manage over 20 services and more than a million lines of code
- I had designed and managed services across agents, harnesses, memory, system engineering, and deployment, ranging in areas like text-to-SQL, text-to-dashboards, forecasting, modeling, and artifact generation and consolidation
- Designed and implemented a hybrid retrieval and search framework along with a text extraction pipeline optimized across several levels of document and query processing for near 1-second retrieval latency
- Designed and implemented a seamless context handoff protocol spanning agents, actions, and workflows

throughout the product

ONGOING RESEARCH PROJECTS

- Scheduler tracing of diffusion language models for understanding and interpretation of characteristics of unmasking schedulers
- Verifiable agentic annotators for multimodal and multiclass datasets through weak supervision and online optimization
- General law of allocation of computation and factorization for sample-efficient learning
- Open-ended generation of unsupervised environments for understanding long-horizon memory and continual learning

ACADEMIC SERVICES

- **Reviewer**, International Conference on Machine Learning 2026, FoGen Workshop
- **Session Chair**, European Chapter of Association of Computational Linguistics 2026
- **Reviewer**, Association of Computational Linguistics 2026, Eval Eval Workshop
- **Reviewer**, International Conference on Learning Representations 2025-2026, DeLTa Workshop
- **Ethics Reviewer**, Neural Information Processing Systems 2025
- **Reviewer**, International Conference on Learning Representations 2025, FM-Wild Workshop
- **Mentor**, Association of Computational Linguistics 2025, Student Research Workshop
- **Speaker**, Unify (YC W23) Research Week

CONFERENCES AND SCHOOLS

- **Attendee**, Y Combinator Startup School India 2026
- **Attendee**, Microsoft Research Academic Summit India 2025
- **Attendee**, IndoML 2025
- **Attendee**, NVIDIA AI Summit 2024
- **Attendee**, ACM Summer School on High Performance Computing in Artificial Intelligence 2023

SKILLS

Deep Learning Tools	PyTorch, PyTorch-Lightning, DeepSpeed, Hugging Face, TensorFlow, vLLM, weights and biases, Langraph, Langfuse
Programming Languages	C++, C, Python, Java, SQL, Bash, L ^A T _E X, Verilog
GPU/cloud Platforms	Modal, Amazon EC2, OpenMPI, Slurm, DigitalOcean
Deployment tools	Docker, Kubernetes, Nginx, CI/CD, Infisical
Web Programming	Django, Flask, JavaScript, FastAPI, LitServe
Databases	MySQL, PostgreSQL, SQLite, Qdrant, Neo4j
Other tools	Supabase, Clickhouse, Redis, Postfix, GoTrue Auth
General Tools	Git, Figma, Jupyterlab/Colab
IDE	Cursor, Windsurf, Warp

RELEVANT COURSES

AI-relevant courses: Machine Learning • Deep Learning • Advanced Machine Learning • Safety Fundamentals of Generative AI • Advanced Topics in Speech Processing • Artificial Intelligence • AI and Ethics • Advanced Digital Image Processing and Computer Vision • Natural Language Processing • Information Retrieval • Applications of Machine Learning in Biological Systems • Computational Biophysics • Stochastic Processes • Probability and Statistics

Computer Science courses: High Performance Computing Architecture • Object Oriented Systems • Advanced Algorithms • Operating Systems • Computer Networks • Database Management Systems • Compilers • Computer Organization and Architecture • Data Structures & Algorithms • Software Engineering • Formal Language and Automata Theory • Discrete Structures

AWARDS AND ACHIEVEMENTS

- I received AMD GPU Open Research Grant for my master's thesis project under the supervision of Prof. Pawan Goyal
- Chosen as part of the top 1% of startups for the Off-Season 2 accelerator by Founders Inc.
- Selected as one of the 15 fellows among 600+ applicants for the Activate AI fellowship
- I was awarded the special mention honors by IIT Kharagpur for extraordinary contributions to technology
- I was awarded a travel grant for attending EACL 2026
- Selected as one of 2,000 participants out of 25,000 applicants for Y Combinator Startup School 2026 and granted \$25k in credits
- IndiaAI Fellowship (2025 & 2026): One of fewer than 200 undergraduate students across India to have twice received the IndiaAI Fellowship from the Government of India, in recognition of innovative contributions to AI research
- AGNI UG Fellowship: awarded to fewer than five undergraduate students, enabling them to engage in advanced research in Artificial Intelligence
- Awarded with a Silver Honour in the International Astronomy and Astrophysics Competition among participants from more than 70 countries
- Secured an ALL INDIA Rank of 1564 in Joint Entrance Examination Advanced among 300,000+ candidates

SOFTWARE PROJECTS

End-to-End Communication Protocol [code](#)

- Designed and implemented My Transport Protocol (MTP) over unreliable UDP sockets, using window-based flow control
- Managed socket communication with multi-threading and shared memory for timeout management, and garbage collection
- Simulated unreliable network links and measured protocol performance under varying packet loss probabilities and timeout values

Foothread Library [code](#)

- Implemented a custom thread-library for process synchronization using existing libraries of semaphores and clone()
- The library has pthread Mutexes and Barriers to manage Mutual exclusion and avoid deadlock with successful Context Switching
- Tested the functioning of library for tree based sum computation ensuring inter-thread synchronization

College Fest Management System [code](#)

- Created a user friendly college festival management website using Node.js, Express.js, and React.js frameworks
- Devised Role - based functionalities for Students, administrators, Volunteers, Organizers, and Event Participants
- Utilized PostgreSQL database with Triggers, Forms, and Schemas for enhanced backend management and data integrity

LLM SQL [code](#)

- Developed an LLM-powered tool to generate SQL queries from natural language, leveraging SQLCoder for precise schema-specific queries
- implemented PostgreSQL server integration with real-time query execution, error handling, and interactive user feedback
- Created a lightweight Flask interface for schema input, query generation, and execution, ensuring efficient data retrieval

OTHER PROJECTS

Safety Decoding in LLMs through Model Merging [code](#)

- Developed a RESTA (Representation-based Safety Transformation and Alignment) based safety decoding pipeline for LLMs with supervised fine-tuning on CodeAlpaca-20k
- Applied DARE (Bernoulli drop & re-scale of SFT deltas) to derive a safety vector from toxic datasets, and merged these vectors with mergekit to produce SFT+RESTA model variants
- Designed and ran safety evaluations on all model variants with HarmEval, using Llama-2-7b-chat as an automated judge for unsafety scores, and assessed held-out test performance with ROUGE-L, METEOR, and BLEU

Mini-copilot - Token Level Python Code Completion [code](#)

- Built a token-level Python code completion system using the CodeXGLUE dataset, including vocabulary curation, sequence normalization, and PyTorch DataLoader pipelines
- Implemented and benchmarked baseline RNN and LSTM architectures, evaluating performance using Top-5 accuracy and perplexity
- Improved the top baseline using a residual recurrent architecture with inter-layer skip connections, and compared accuracy, perplexity, and parameter efficiency

Image Captioning Encoder-Decoder Architecture [code](#)

- Implemented and trained image captioning models, including CNN+LSTM utilizing ResNet-50 as encoder and ViT + GPT-2 architecture
- Built the vocabulary and tokenizer from scratch for CNN + LSTM architecture and achieved a ROUGE-L score of 0.27 on the test dataset
- Finetuned ViT + GPT-2 architecture for 3 epochs on our dataset using LORA and achieved a ROUGE-L score of 0.34 on the test dataset

PORTFOLIOS

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- X/Twitter: <https://x.com/devichand579>
- Github: <https://github.com/devichand579>
- Website: <https://devichand.com>

Other Publications

- Devichand Budagam, Azamat Zhanatuly Imanbayev, Iskander Akhmetov, Aleksandr M. Sinitca, Sergey Antonov, Dmitrii Kaplun. “A Panoramic Dental X-ray Dataset for Instance Segmentation and Teeth Numbering” **Preprint** [PDF]